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DeepFake Video Detection System

By Group 10



INTRODUCTION

1. Deepfakes are AI-generated media that manipulate visual or audio content.
2. They are created using advanced machine learning techniques.
3. An AI system first needs to learn the facial features, expressions, and mannerisms of a real person from a large dataset of their images and videos.
4. The system analyzes the movements, speech patterns, and tone of voice of the person whose face is to be superimposed onto another's body.
5. Once the AI system has learned these features, it can generate a fake video that looks and sounds like the target person is saying or doing something that they never did in reality.
6. Deepfakes have the potential to be used for malicious purposes.
7. Detecting and combating deepfakes has become a critical challenge in the digital age.



NBC News

The deepfake AI porn industry is operating in plain sight

Deepfake porn made using AI software that takes an existing video and seamlessly replace one person's face with another's has taken off...

1 month ago



Creative Bloq

Deepfake cases growing everyday

These examples of the best deepfake examples prove how far technology has come, for better or for worse.

1 month ago



Fast Company

AI celebrity deepfakes are morphing into hoaxes as entertainment

The Balenciaga pope was just the beginning as Biden, Trump, and Rogan deepfakes show how AI is blurring the line between political cartoon...

3 weeks ago



Forbes

Republicans Share An Apocalyptic AI-Powered Attack Ad Against Biden: Here's How To Spot A Deepfake

Deepfakes first came about in 2017 after a Reddit user posted pornography with the adult actors' faces swapped for the faces of celebrities.

5 days ago

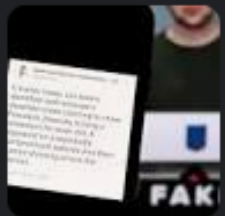


Bloomberg Law News

AI Celebrity 'Deepfakes' Clash With Web of State Publicity Laws

The swift rise of "deepfakes" and similar artificial intelligence technology that allows users to digitally swap or blend faces has spurred...

2 weeks ago



Little Black Book

What's the Law on Deepfake Celebs in Ads and Entertainment? | LBBOnline



THE PURPOSE

1. Growing concern about deepfakes due to their potential harm to individuals and society.
2. Deepfakes are AI-generated media that manipulate visual or audio content to depict someone doing or saying something they never did in reality.
3. Detecting and combating deepfakes has become a critical challenge in the digital age.
4. The purpose of the project is to develop a deepfake detection system that accurately identifies fake videos created using machine learning techniques.

Problem Statement

The increasing sophistication of smartphone cameras and the availability of good internet connections all over the world has increased the ever-growing reach of social media and media sharing portals have made the creation and transmission of digital videos easier than ever before. The growing computational power has made deep learning so powerful that would have been thought impossible only a handful of years ago.

Like any transformative technology, this has created new challenges. So-called "DeepFake" is produced by deep generative adversarial models that can manipulate video and audio clips. Spreading of the DF over the social media platforms has become very common leading to spamming and peculating wrong information over the platform. These types of DF will be terrible and lead to threatening, misleading common people.

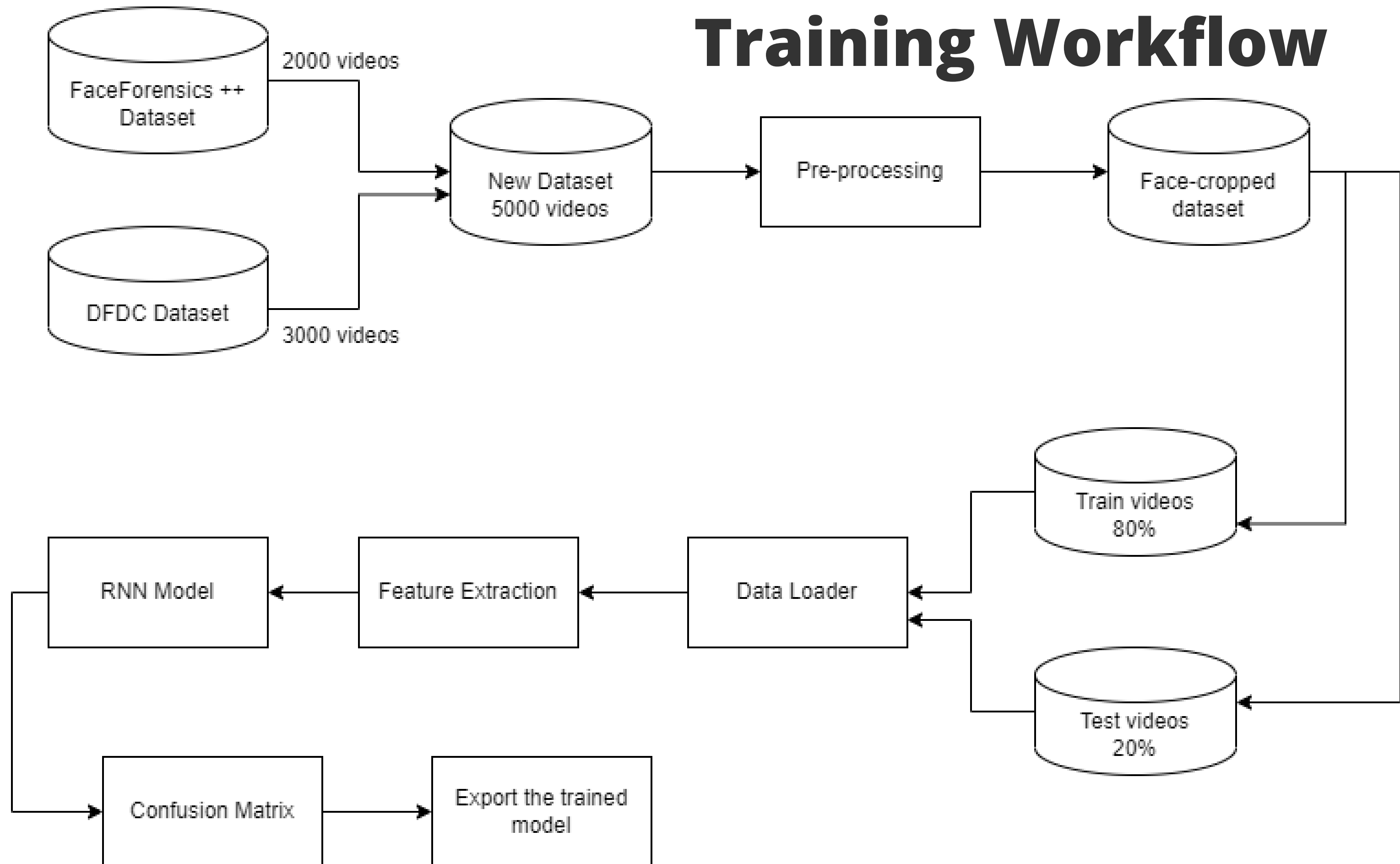


Dataset Exploration

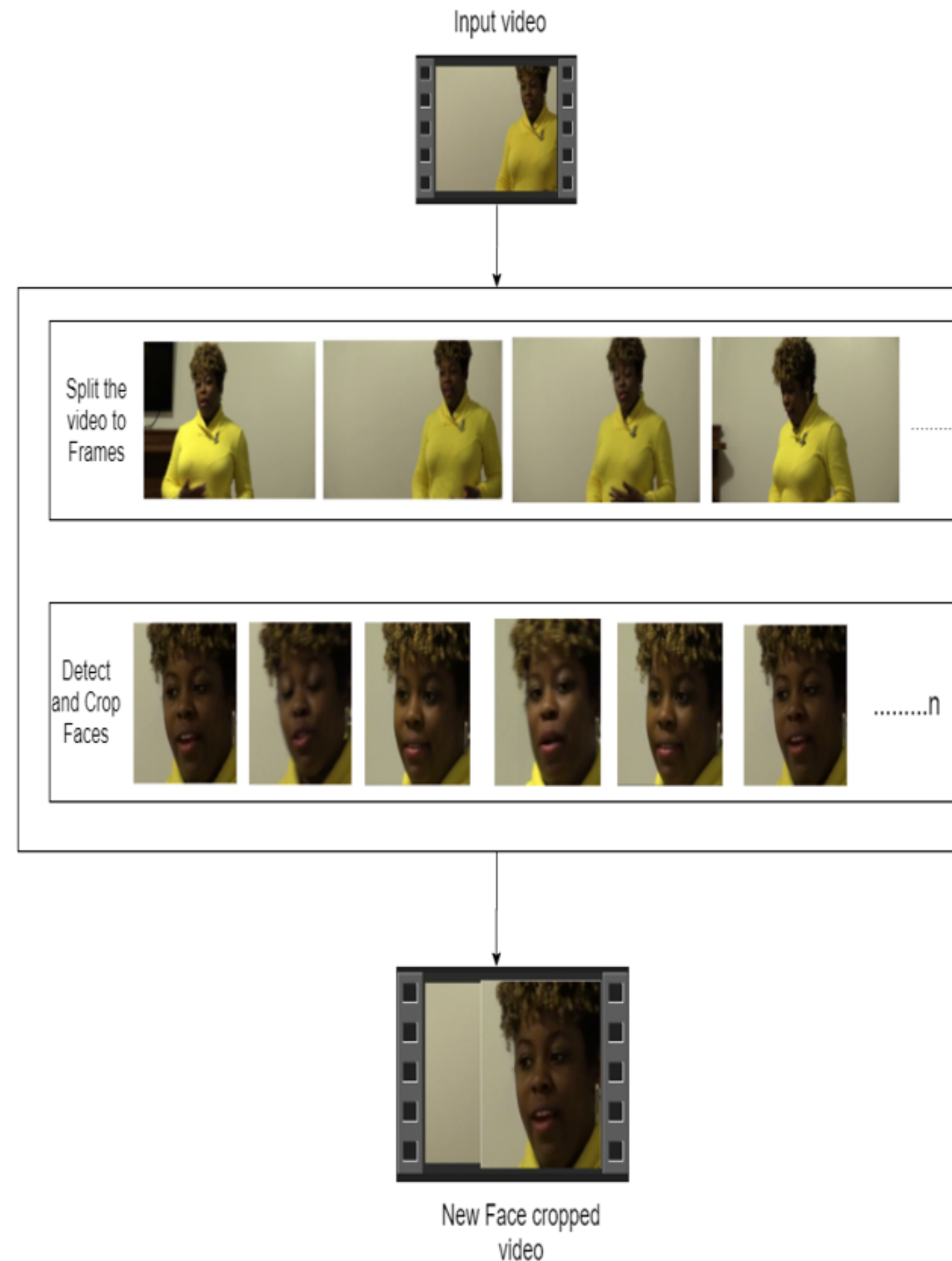
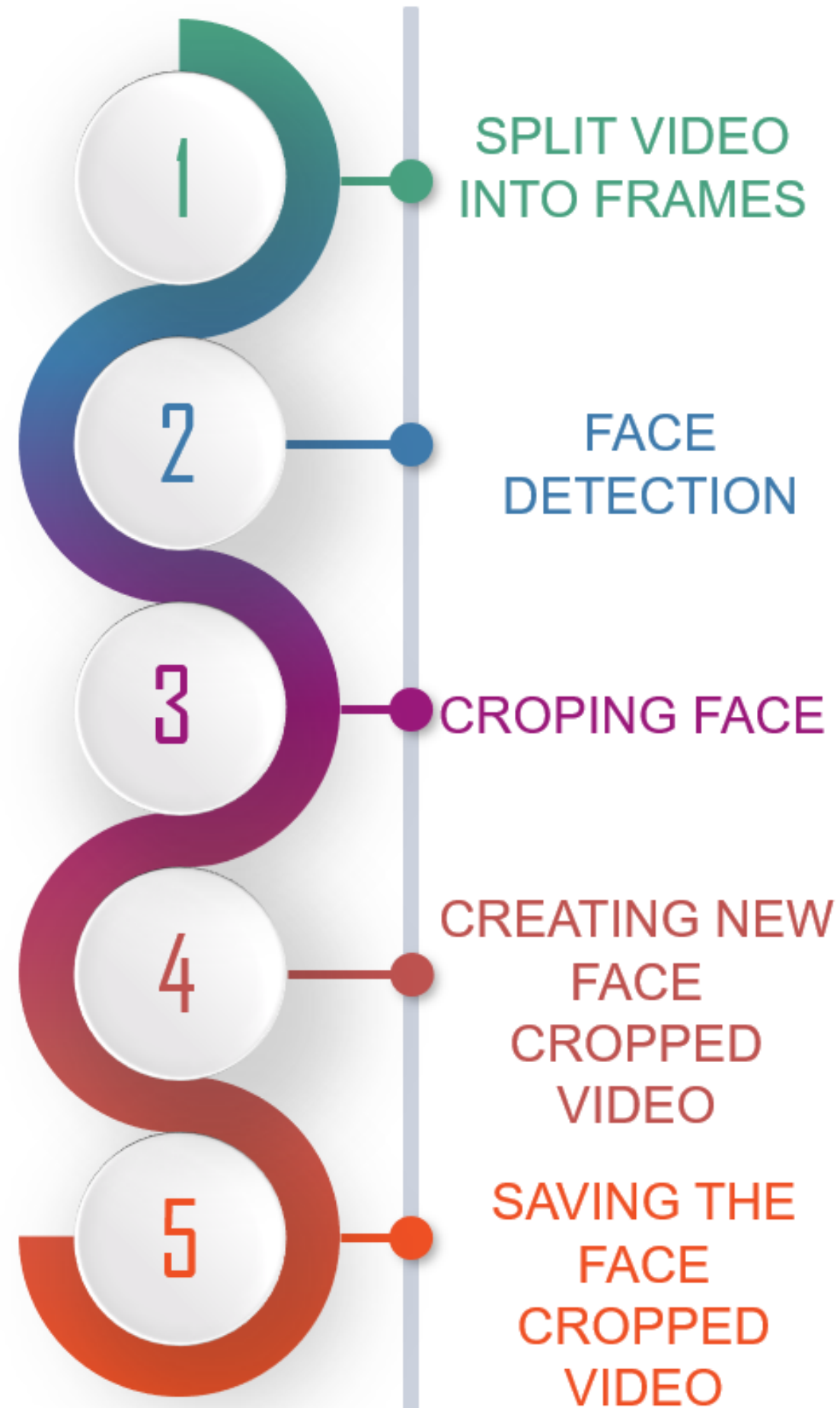
Dataset from two different sources –

- DeepFake detection challenge (Kaggle)
- 50% real and 50% fake video to avoid bias.
- Removed audio altered DeepFakes from DFDC.
- Dataset is then passed to pre-processing phase.

Training Workflow



Pre-processing



Face Detection Algorithm

1. Created a balanced dataset of 50% real and 50% fake videos
2. Used OpenCV for face extraction
3. Determined 150 frames for each video for consistency
4. Applied Haar Cascade Classifier algorithm for face detection
5. Extracted and resized the face region to 112x112 pixels
6. Used VideoWriter function to create a new video file

Accuracies of other models

Title of Paper	Techniques Used	Dataset used	Accuracy
Deepfake Video Detection through Optical Flow based CNN[	Convolutional Neural Network (CNN)	Face2Face	VGG16 81.61%, ResNet50 75.46%
FaceForensics++: Learning to Detect Manipulated Facial Images.	Xception net (CNN)	FaceForensics++	81%
Multi-scale face detection based on a convolutional neural network.[14]	Convolution Neural Networks (CNN)	1. CelebA 2. AFW 3. FDDB	Discrete- 90% and for continuous, it is 74%
Exploiting Visual Artifacts to Expose Deep Fakes and Face Manipulations	The neural network classifier as MLP and the logistic regression model as LogReg	1.CelebA 2.ProGAN 3.Glow	MLP 84%(Eyes), LogReg 83% (Eyes)

Model Architecture

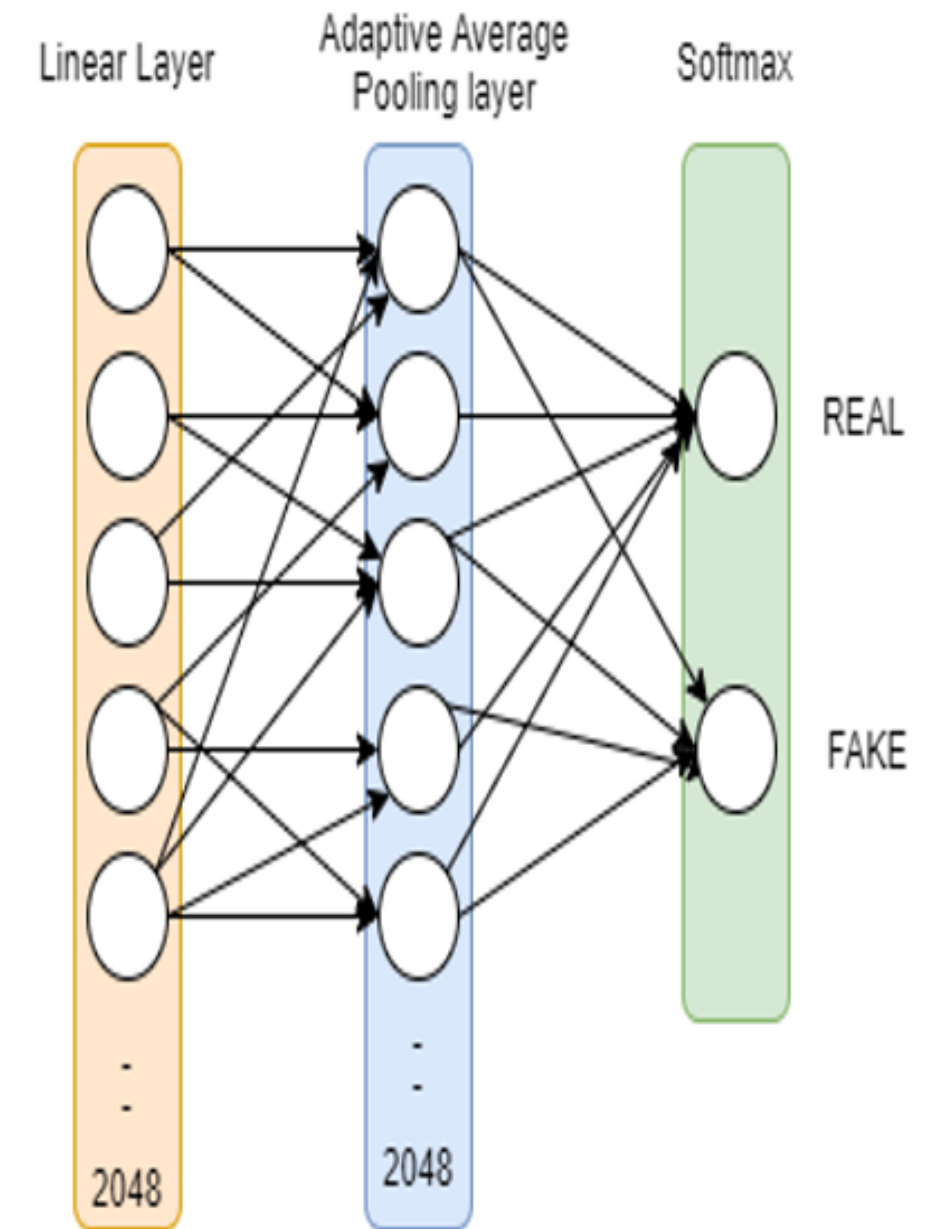
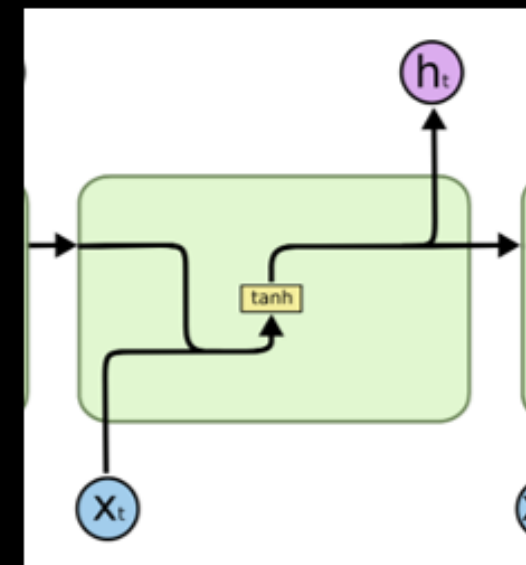


ResNext-50

stage	output	ResNeXt-50 (32×4d)
conv1	112×112	7×7, 64, stride 2
conv2	56×56	3×3 max pool, stride 2
		$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128, C=32 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3	28×28	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256, C=32 \\ 1 \times 1, 512 \end{bmatrix} \times 4$
conv4	14×14	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512, C=32 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$
conv5	7×7	$\begin{bmatrix} 1 \times 1, 1024 \\ 3 \times 3, 1024, C=32 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	global average pool 1000-d fc, softmax
# params.		25.0×10^6

Sequential Layer

1 LSTM layer with
2048 shape input
vector and 2048
latent features along
with 0.4
chance of dropout
and ReLU
Activation function



Questions Regarding ResNeXt Layer

Q1. What is resnext50_32x4d?

Q2. On what factors this model is predicting whether a video is fake or real?

Q3. So is res next detecting statistical differences too or is it just for feature extraction?

Q4. What are the statistical differences in the DF videos?

Q5. Why is LSTM not used directly without including res next?

Q6. How does resnext which is trained on imagenet dataset is able to detect statistical differences?

Q7. Why is resnext used as a CNN and why not other CNN like resnet or deepnet etc..

Questions Regarding LSTM Layer

Q1. Why is LSTM used as an RNN?

Q2. Why not other RNNs?

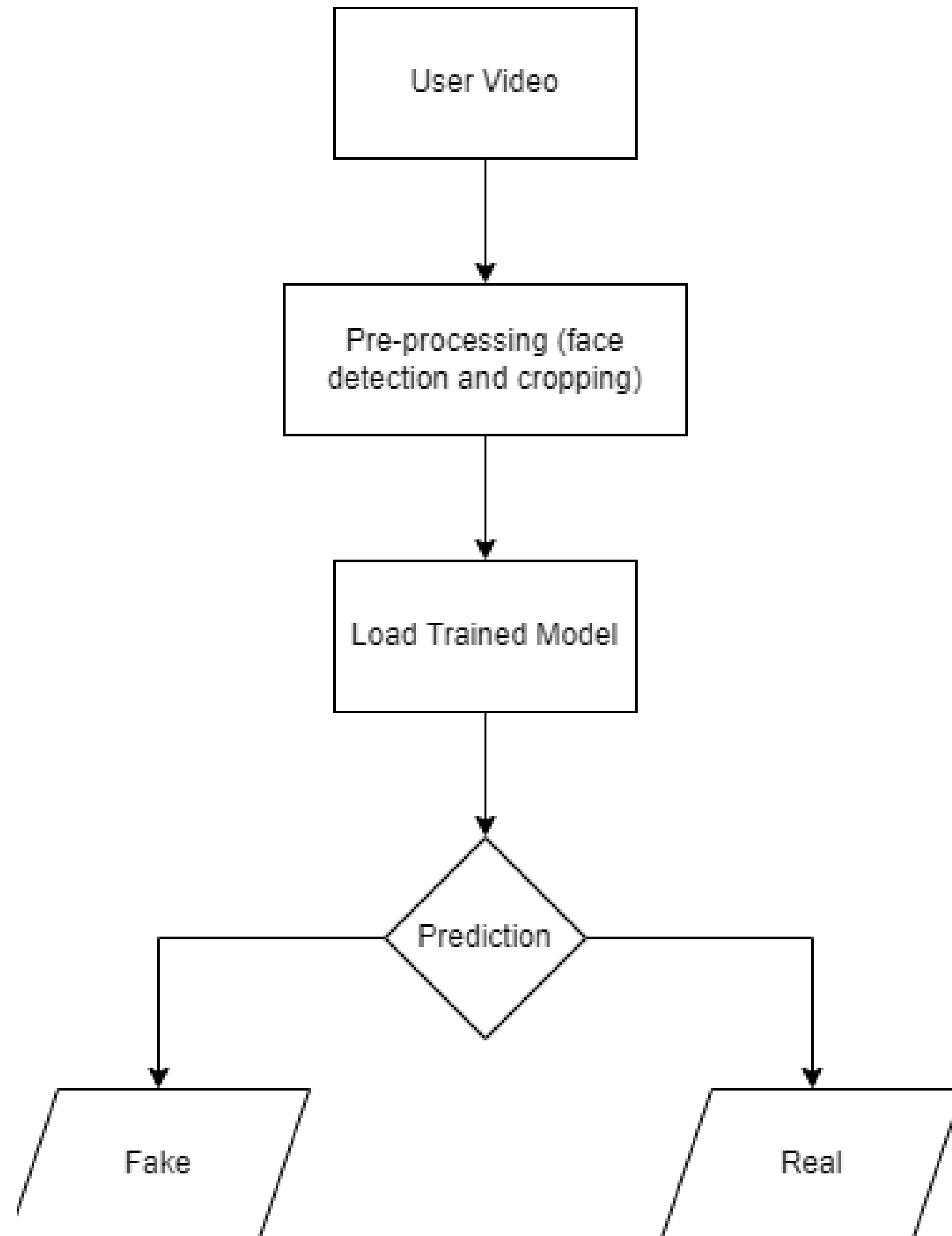
Q3. What are those temporal dependencies?

Q4. How a deepfake generated videos have temporal inconsistencies?

Q5. So is this model only capturing temporal dependencies or it is capturing other things too?

Q6. So given a video, explain in detail how this model will detect if it is deepfake or not?

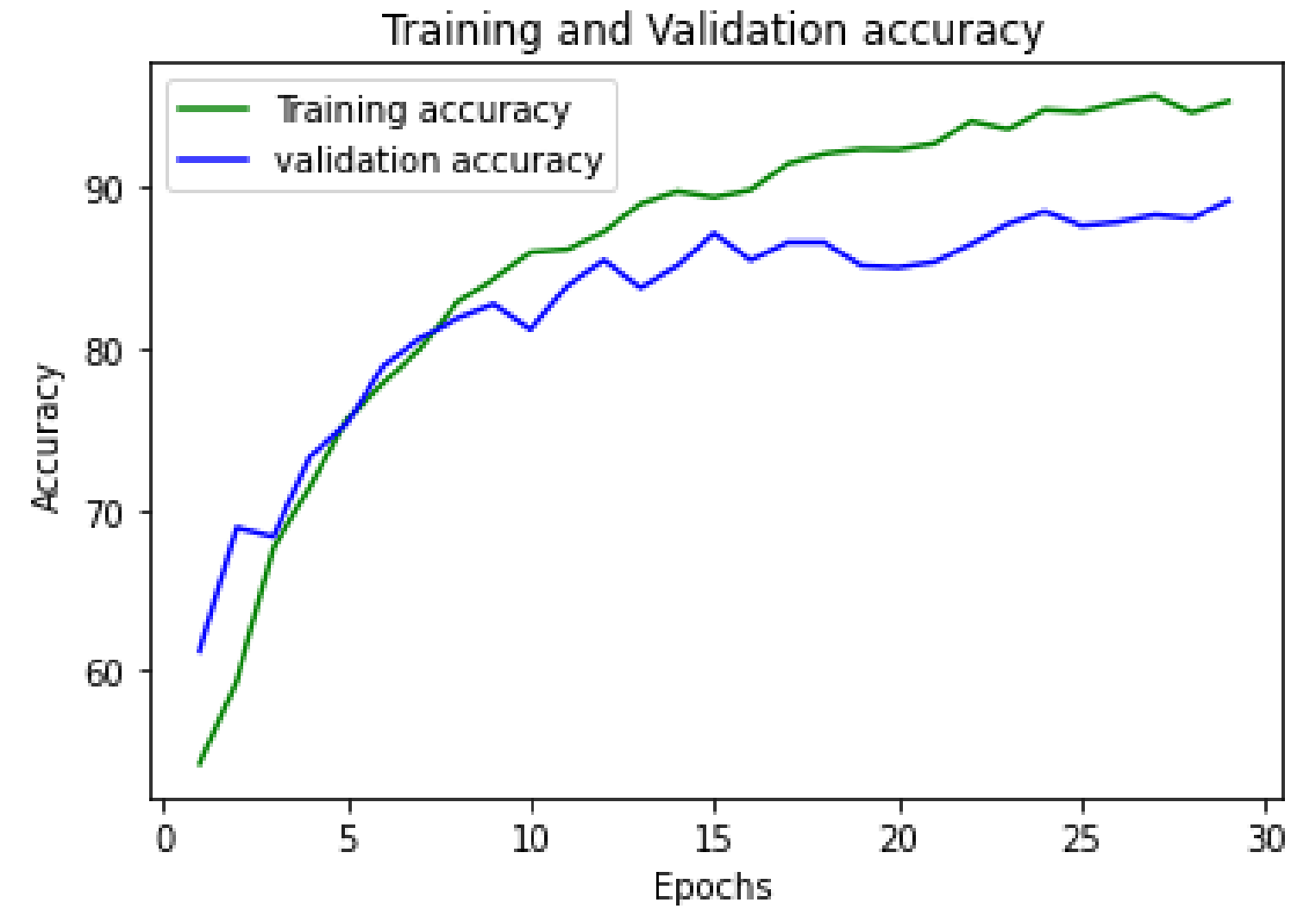
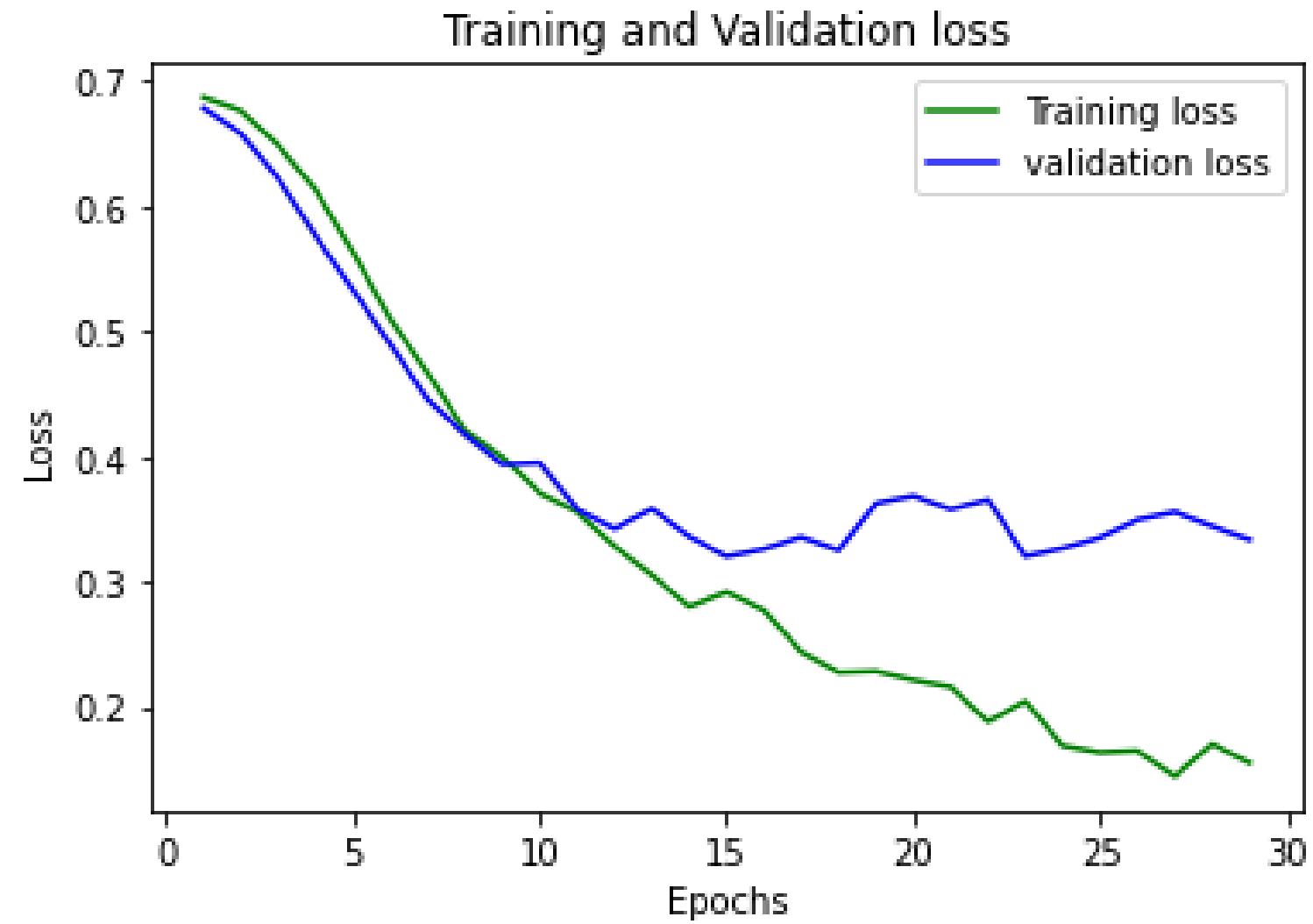
Prediction Workflow



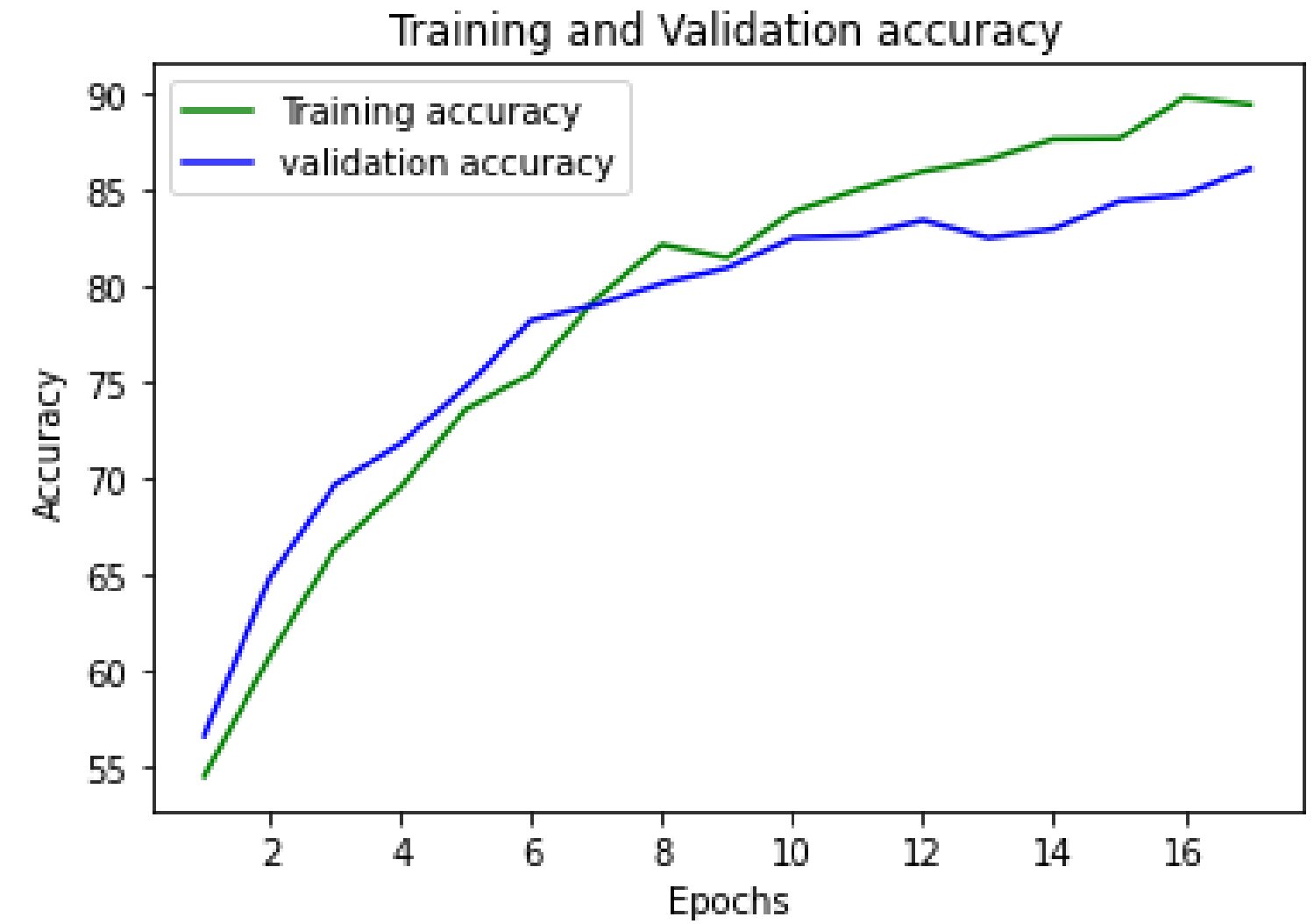
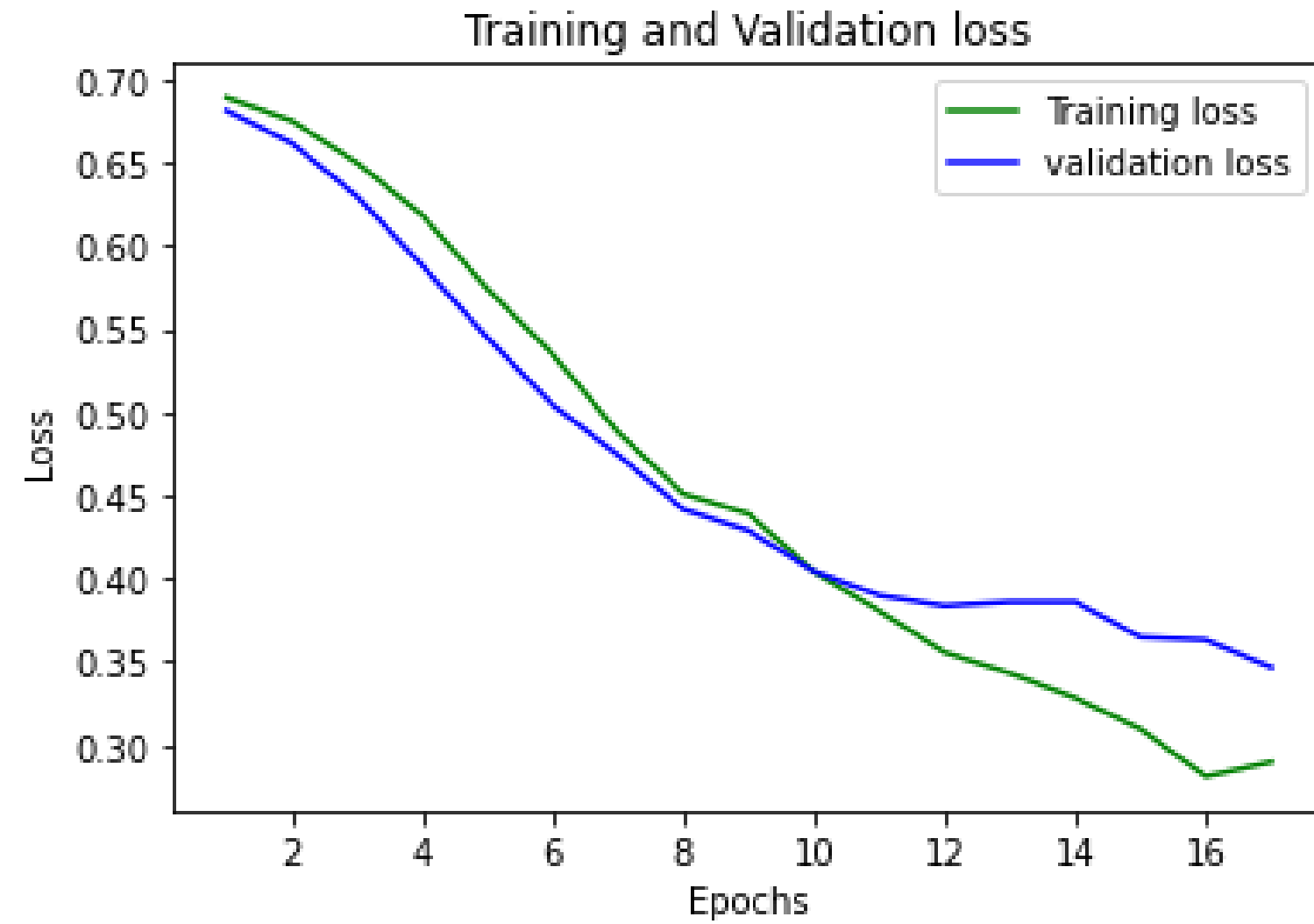
Results

Model Name	Sequence Length	Accuracry
model_10_frames	10	83.22072072072072
model_20_frames	20	86.03603603603604
model_40_frames	40	89.1891891891892
model_60_frames	60	91.8271085934174

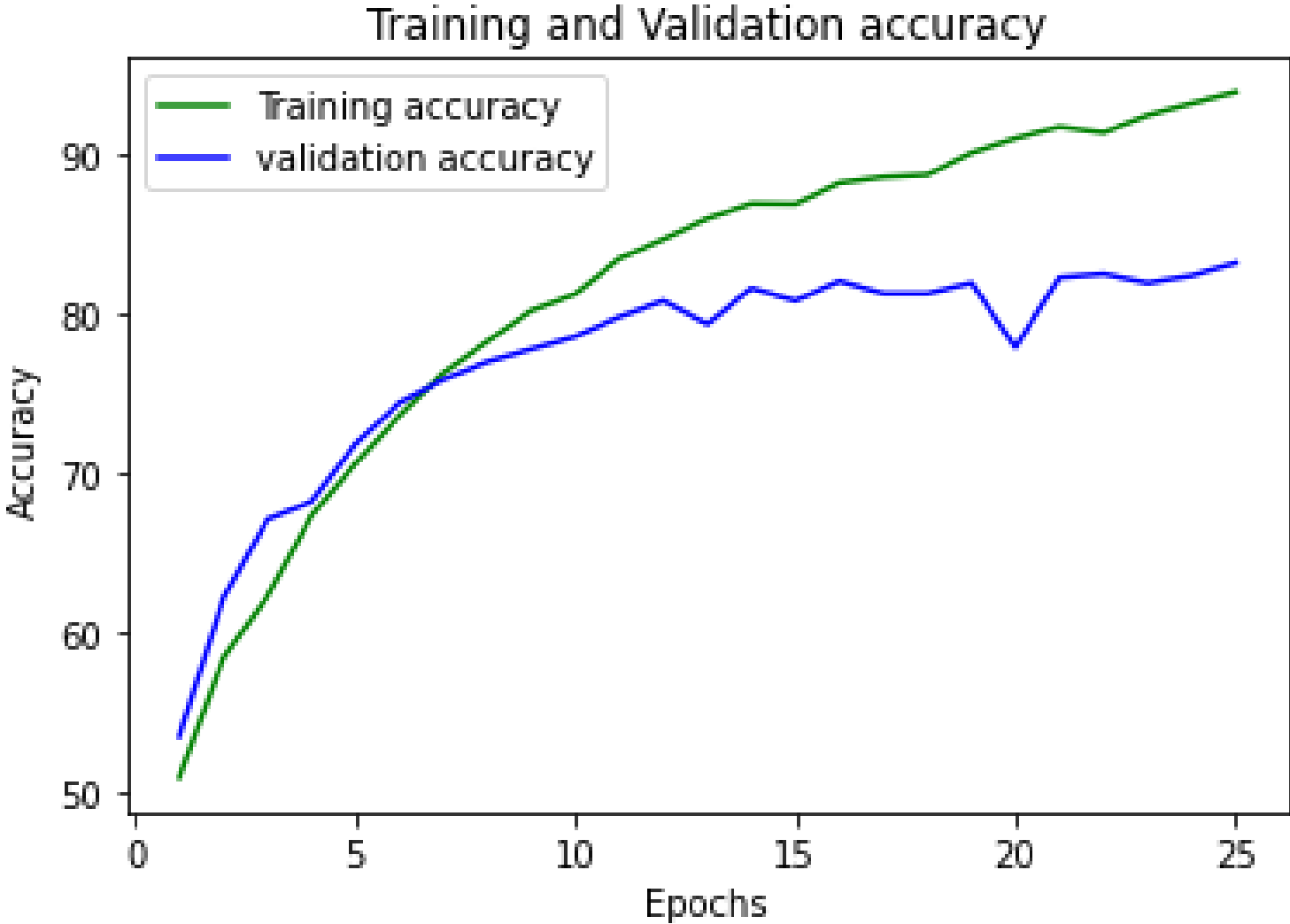
model_40_frames



model_20_frames



model_10_frames



Tools and Technologies



Programming Languages

- Python3
- JavaScript



Programming Frameworks

- PyTorch
- Django

django



IDE

- Google **colab**
- Jupyter Notebook
- Visual Studio Code



Cloud Services

- Google Cloud Platform



Version Control

- Git

Thank you!

